

D361

# Business Simulation

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

In the simulation your company sells a fitness tracker for \$80. Variable cost is \$50 per unit, and fixed costs for the round are \$450,000. A rival just cut its price to \$75, and your ad agency won an award. How many units must you sell to break even this round?

- A. 5,625
- B. 9,000
- C. 15,000
- D. 18,000

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2. [TRANSFER — reversed stem / different variable]

A product sells for \$40 with a variable cost of \$25 and breaks even at 12,000 units. What are its fixed costs?

- A. \$800
- B. \$300,000
- C. \$480,000
- D. \$180,000

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

Your simulation's market report says demand for your mid-range laptop has a price elasticity of  $-2.5$ . Your plant is running at 70% capacity and your brand ranks second in awareness. You are thinking of cutting price 4%. What happens to total revenue?

- A. It falls, because each unit brings in less.
- B. It stays the same, because price and quantity changes cancel.
- C. It rises, because demand is elastic.
- D. It rises only if the plant stays below 70% capacity.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**4.** [TRANSFER — reversed stem / different variable]

Demand for a product has a price elasticity of  $-0.4$ . To increase total revenue, what should the firm do with price?

- A.** Raise it.
- B.** Keep it; elasticity below 1 means demand cannot change at all.
- C.** Nothing; price changes never affect revenue.
- D.** Cut it.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**5.**

In round 4, your firm sold 36,000 units. The four competitors sold 42,000, 30,000, 24,000 and 18,000 units. Your firm also spent \$2 million on R&D and opened a second plant. What is your market share?

- A.** 85.7%
- B.** 24%
- C.** 31.6%
- D.** 20%

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**6.**

Your simulated firm uses 20,000 units of a component a year. Each order costs \$200 to place, and holding one unit for a year costs \$5. The supplier is 300 miles away and offers free delivery. What is the economic order quantity?

- A.** About 1,265 units
- B.** About 32 units
- C.** About 894 units
- D.** 20,000 units, since delivery is free

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

Your team spent \$3 million in earlier rounds developing a product line that now loses money. Continuing it next round would bring \$1.2 million in revenue and \$1.5 million in costs that can be avoided if you drop it. The line was your CEO's idea, and it won a design award. What should you do?

- A. Keep it; dropping any product always hurts the brand.
- B. Keep it to earn back the \$3 million.
- C. Keep it because it won a design award.
- D. Drop it; the \$3 million is sunk and continuing loses \$300,000 more.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8. [TRANSFER — reversed stem / different variable]

A firm is deciding whether to accept a one-time special order. Which cost is relevant?

- A. Rent paid whether or not the order is accepted.
- B. Last year's cost of building the factory.
- C. Every cost on the income statement, since all costs matter.
- D. Extra materials needed only for this order.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9.

You can fund one project. Project A costs \$400,000 and adds \$100,000 profit a year. Project B costs \$1,000,000 and adds \$180,000 a year. Project B would be built by your preferred contractor. Based on annual ROI, which is better?

- A. B, because of the preferred contractor
- B. A, at 25%
- C. A, at 4.0
- D. B, because \$180,000 is larger

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**10.**

Your plant can make 50,000 units per round. Last round it produced 42,000 units and sold 39,000. Your marketing team also redesigned the packaging. What was capacity utilization?

- A.** 119%
- B.** 84%
- C.** 93%
- D.** 78%

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**11.**

Your firm buys a product for \$60 and sells it for \$80. The product ships from a warehouse in Ohio and is rated four stars by customers. What is the gross margin percentage?

- A.** 25%
- B.** \$20
- C.** 33.3%
- D.** 75%

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**12.** [TRANSFER — reversed stem / different variable]

A retailer wants a 40% gross margin on an item that costs \$30. What price should it set?

- A.** \$75
- B.** \$12
- C.** \$50
- D.** \$42

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer C Concept 1

In the simulation your company sells a fitness tracker for \$80. Variable cost is \$50 per unit, and fixed costs for the round are \$450,000. ~~A rival just cut its price to \$75, and your ad agency won an award.~~ How many units must you sell to break even this round?

**Target word:** break-even units **Decides:** Contribution margin =  $\$80 - \$50 = \$30$ .

|   |                    |                                                          |
|---|--------------------|----------------------------------------------------------|
| A | Dropped element    | Divides by price; drops variable cost.                   |
| B | Inverted operation | Divides by variable cost instead of contribution margin. |
| C | <b>Correct</b>     | $\$450,000 \div \$30 = 15,000$ .                         |
| D | Camouflage         | Uses the rival's \$75 price: $\$450,000 \div \$25$ .     |

## 2. Answer D Concept 1 • transfer

A product sells for \$40 with a variable cost of \$25 and breaks even at 12,000 units. What are its fixed costs?

**Target word:** solve for fixed costs **Decides:** Fixed costs = break-even units  $\times$  contribution margin.

|   |                    |                                                |
|---|--------------------|------------------------------------------------|
| A | Inverted operation | Divides units by margin.                       |
| B | Wrong stem number  | Uses variable cost per unit instead of margin. |
| C | Dropped element    | Uses price and drops variable cost.            |
| D | <b>Correct</b>     | $12,000 \times (\$40 - \$25) = \$180,000$ .    |

## 3. Answer C Concept 2

Your simulation's market report says demand for your mid-range laptop has a price elasticity of  $-2.5$ . ~~Your plant is running at 70% capacity and your brand ranks second in awareness.~~ You are thinking of cutting price 4%. What happens to total revenue?

**Target word:** direction of revenue **Decides:**  $|E| = 2.5 > 1$ : quantity rises about 10% for a 4% cut.

|   |                    |                                                             |
|---|--------------------|-------------------------------------------------------------|
| A | Inverted operation | Ignores the larger rise in quantity.                        |
| B | Adjacent category  | That is unit-elastic demand ( $ E  = 1$ ).                  |
| C | <b>Correct</b>     | $0.96 \times 1.10 \approx 1.056$ , about 5.6% more revenue. |
| D | Camouflage         | Capacity use does not decide the elasticity effect here.    |

## 4. Answer A Concept 2 • transfer

Demand for a product has a price elasticity of  $-0.4$ . To increase total revenue, what should the firm do with price?

**Target word:** price move for inelastic demand **Decides:**  $|E| < 1$ : quantity barely responds.

|   |                    |                                                         |
|---|--------------------|---------------------------------------------------------|
| A | <b>Correct</b>     | Price rises more than quantity falls, so revenue grows. |
| B | Confusion set      | Confuses inelastic with perfectly inelastic.            |
| C | Absolute language  | Revenue changes unless demand is exactly unit elastic.  |
| D | Inverted operation | Right for elastic demand, wrong here.                   |

### 5. Answer B Concept 3

In round 4, your firm sold 36,000 units. The four competitors sold 42,000, 30,000, 24,000 and 18,000 units. ~~Your firm also spent \$2 million on R&D and opened a second plant.~~ What is your market share?

**Target word:** unit market share **Decides:** Total market = 150,000 units.

|   |                   |                                                           |
|---|-------------------|-----------------------------------------------------------|
| A | Wrong stem number | Divides by the leader's 42,000.                           |
| B | <b>Correct</b>    | $36,000 \div 150,000 = 24\%$ .                            |
| C | Dropped element   | Leaves your own sales out of the total ( $36 \div 114$ ). |
| D | Adjacent category | Assumes five equal shares.                                |

### 6. Answer A Concept 4

Your simulated firm uses 20,000 units of a component a year. Each order costs \$200 to place, and holding one unit for a year costs \$5. ~~The supplier is 300 miles away and offers free delivery.~~ What is the economic order quantity?

**Target word:** EOQ **Decides:**  $\sqrt{(2 \times 20,000 \times 200 \div 5)} = \sqrt{1,600,000}$ .

|   |                    |                                                                     |
|---|--------------------|---------------------------------------------------------------------|
| A | <b>Correct</b>     | $\sqrt{1,600,000} \approx 1,265$ .                                  |
| B | Inverted operation | Swaps order cost and holding cost.                                  |
| C | Dropped element    | Drops the 2 in the formula.                                         |
| D | Camouflage         | Free delivery does not remove the \$200 order cost or holding cost. |

### 7. Answer D Concept 5

Your team spent \$3 million in earlier rounds developing a product line that now loses money. Continuing it next round would bring \$1.2 million in revenue and \$1.5 million in costs that can be avoided if you drop it. ~~The line was your CEO's idea, and it won a design award.~~ What should you do?

**Target word:** keep or drop **Decides:** Future: \$1.2M revenue vs. \$1.5M avoidable cost.

|   |                   |                                                |
|---|-------------------|------------------------------------------------|
| A | Absolute language | No rule says every drop hurts.                 |
| B | Wrong stem number | Uses the sunk cost, which cannot be recovered. |
| C | Camouflage        | Awards do not change the cash math.            |
| D | <b>Correct</b>    | Only future, avoidable amounts matter.         |

### 8. Answer D Concept 5 · transfer

A firm is deciding whether to accept a one-time special order. Which cost is relevant?

**Target word:** pick the relevant cost **Decides:** Relevant = future and differs between choices.

|   |                   |                                                  |
|---|-------------------|--------------------------------------------------|
| A | Adjacent category | Fixed and unchanged by the decision.             |
| B | Confusion set     | A sunk cost.                                     |
| C | Absolute language | Only costs that change with the decision matter. |
| D | <b>Correct</b>    | Incurred only if the order is accepted.          |

### 9. Answer B Concept 6

You can fund one project. Project A costs \$400,000 and adds \$100,000 profit a year. Project B costs \$1,000,000 and adds \$180,000 a year. ~~Project B would be built by your preferred contractor.~~ Based on annual ROI, which is better?

**Target word:** higher ROI **Decides:** A = 25%, B = 18%.

|          |                    |                                              |
|----------|--------------------|----------------------------------------------|
| A        | Camouflage         | The contractor does not change ROI.          |
| <b>B</b> | <b>Correct</b>     | $\$100,000 \div \$400,000 = 25\%$ beats 18%. |
| C        | Inverted operation | Divides cost by profit.                      |
| D        | Dropped element    | Ignores the investment size.                 |

### 10. Answer B Concept 7

Your plant can make 50,000 units per round. Last round it produced 42,000 units and sold 39,000. ~~Your marketing team also redesigned the packaging.~~ What was capacity utilization?

**Target word:** utilization **Decides:** Production  $\div$  capacity.

|          |                    |                                                           |
|----------|--------------------|-----------------------------------------------------------|
| A        | Inverted operation | Capacity $\div$ output.                                   |
| <b>B</b> | <b>Correct</b>     | $42,000 \div 50,000 = 84\%$ .                             |
| C        | Confusion set      | Sales $\div$ production is sell-through, not utilization. |
| D        | Adjacent category  | Uses units sold, not produced.                            |

### 11. Answer A Concept 8

Your firm buys a product for \$60 and sells it for \$80. ~~The product ships from a warehouse in Ohio and is rated four stars by customers.~~ What is the gross margin percentage?

**Target word:** gross margin % **Decides:** Divide the \$20 gross profit by price.

|          |                    |                                        |
|----------|--------------------|----------------------------------------|
| <b>A</b> | <b>Correct</b>     | $\$20 \div \$80 = 25\%$ .              |
| B        | Adjacent category  | Dollar gross profit, not a percentage. |
| C        | Confusion set      | That is markup: $\$20 \div \$60$ .     |
| D        | Inverted operation | Cost $\div$ price, the cost ratio.     |

### 12. Answer C Concept 8 • transfer

A retailer wants a 40% gross margin on an item that costs \$30. What price should it set?

**Target word:** solve for price **Decides:** Price = cost  $\div$  (1 – margin).

|          |                    |                                                            |
|----------|--------------------|------------------------------------------------------------|
| A        | Inverted operation | Divides cost by the margin instead of by 1 – margin.       |
| B        | Wrong stem number  | 40% of cost, not a price.                                  |
| <b>C</b> | <b>Correct</b>     | $\$30 \div 0.60 = \$50$ ; margin $\$20 \div \$50 = 40\%$ . |
| D        | Confusion set      | A 40% markup on cost, which is only a 28.6% margin.        |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 5             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 4             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 5             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 9             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 3             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 5             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 5             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                           | Rule                                                                                                                              | Trap                                                               | Tell                                                                  |
|---|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1 | <b>Contribution margin and break-even</b>         | Break-even units = fixed costs ÷ (price – variable cost per unit).                                                                | Dividing by price, or using a rival's price.                       | Your price, your unit variable cost and your fixed costs in one stem. |
| 2 | <b>Price elasticity and revenue</b>               | When demand is elastic ( $ E  > 1$ ), a price cut raises total revenue; when inelastic ( $ E  < 1$ ), a price increase raises it. | Assuming lower prices always mean lower (or higher) revenue.       | An elasticity figure next to a planned price change.                  |
| 3 | <b>Market share</b>                               | Market share = your sales ÷ total market sales, including your own.                                                               | Leaving your own sales out of the total, or assuming equal shares. | Your units plus each rival's units.                                   |
| 4 | <b>Economic order quantity</b>                    | $EOQ = \sqrt{(2 \times \text{annual demand} \times \text{order cost} \div \text{annual holding cost per unit})}$ .                | Dropping the 2, or swapping order and holding costs.               | Annual demand, cost per order and holding cost per unit.              |
| 5 | <b>Sunk costs are irrelevant</b>                  | Decide on future, avoidable costs and revenues only; money already spent cannot be recovered.                                     | Continuing a loser to recoup past spending.                        | A large past investment next to a choice about the future.            |
| 6 | <b>Return on investment for choosing projects</b> | ROI = annual profit ÷ investment; compare rates, not dollar profits, when sizes differ.                                           | Picking the project with the bigger dollar profit.                 | Two projects of different sizes.                                      |
| 7 | <b>Capacity utilization</b>                       | Utilization = actual output ÷ capacity.                                                                                           | Using units sold instead of units produced.                        | Capacity, production and sales all given.                             |
| 8 | <b>Gross margin vs. markup</b>                    | Gross margin % = (price – cost) ÷ price; markup % = (price – cost) ÷ cost.                                                        | Reporting markup when margin is asked.                             | A cost and a price and the word margin or markup.                     |

**Course scope (public description):** This course ties together all the skills and knowledge covered in the business courses and allows the student to prove their mastery of the competencies by applying them in a simulated business environment. This course will help take the student's knowledge and skills from the theoretical to applicable.